

SHETH MVLU COLLEGE

SUBJECT:- WSN

Aim: Implementation of Events, Commands, and Tasks in TinyOS Develop a TinyOS program that toggles LEDs using events and tasks to demonstrate split-phase execution.

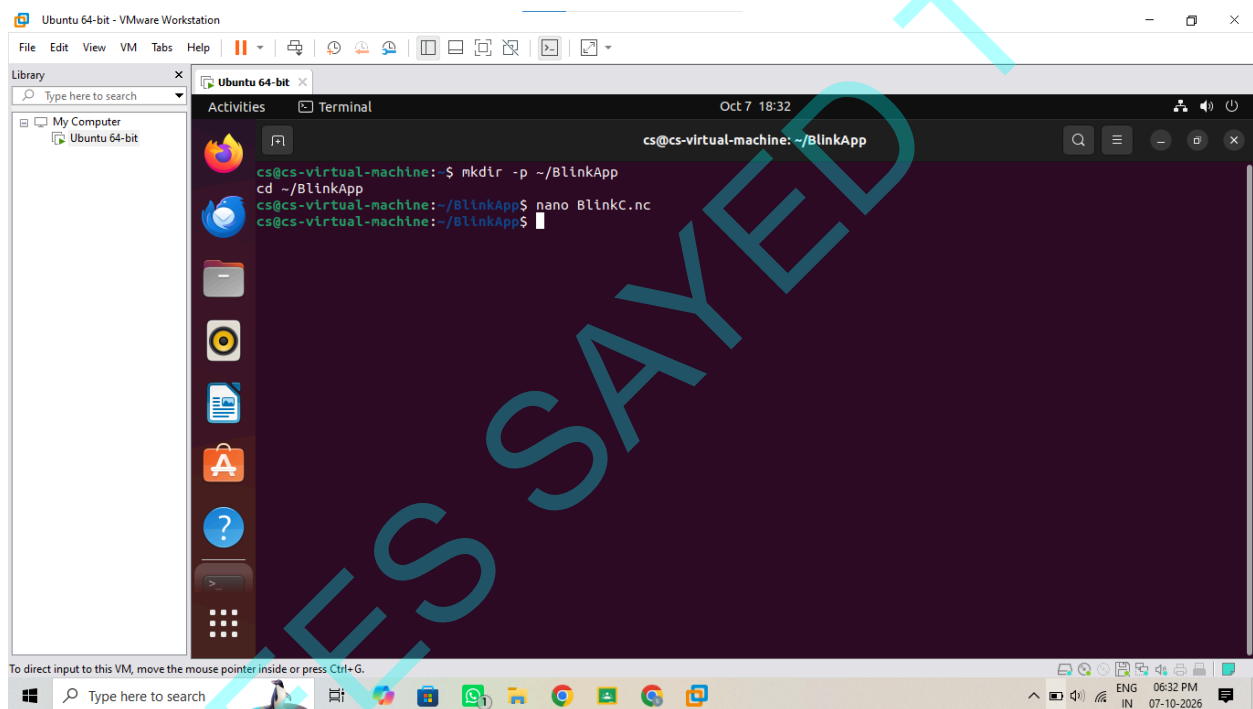
Create a separate folder for the practical:

```
mkdir -p ~/BlinkApp
```

```
cd ~/BlinkApp
```

Create the main nesC module:

```
nano BlinkC.nc
```



```
#include "Timer.h"
```

```
module BlinkC
```

```
{
    uses interface Timer<TMilli> as Timer0;
    uses interface Leds;
    uses interface Boot;
}
```

```
implementation
```

```
{
    uint16_t counter = 0;
```

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```
// TASK: Deferred background execution
task void toggleLedTask()
{
    counter++;

    dbg("BlinkC",
        "TASK RUNNING: Processing split-phase task, counter = %d\n",
        counter);

    // COMMAND: Toggle LED0
    call Leds.led0Toggle();

    // Toggle LED1 on every alternate execution
    if (counter % 2 == 0)
    {
        call Leds.led1Toggle();

        dbg("BlinkC",
            "TASK: LED1 toggled on even count.\n");
    }
}

// EVENT: System booted
event void Boot.booted()
{
    dbg("BlinkC",
        "BOOT EVENT: System booted, starting periodic timer.\n");

    // COMMAND: Start 250 ms periodic timer
    call Timer0.startPeriodic(250);
}

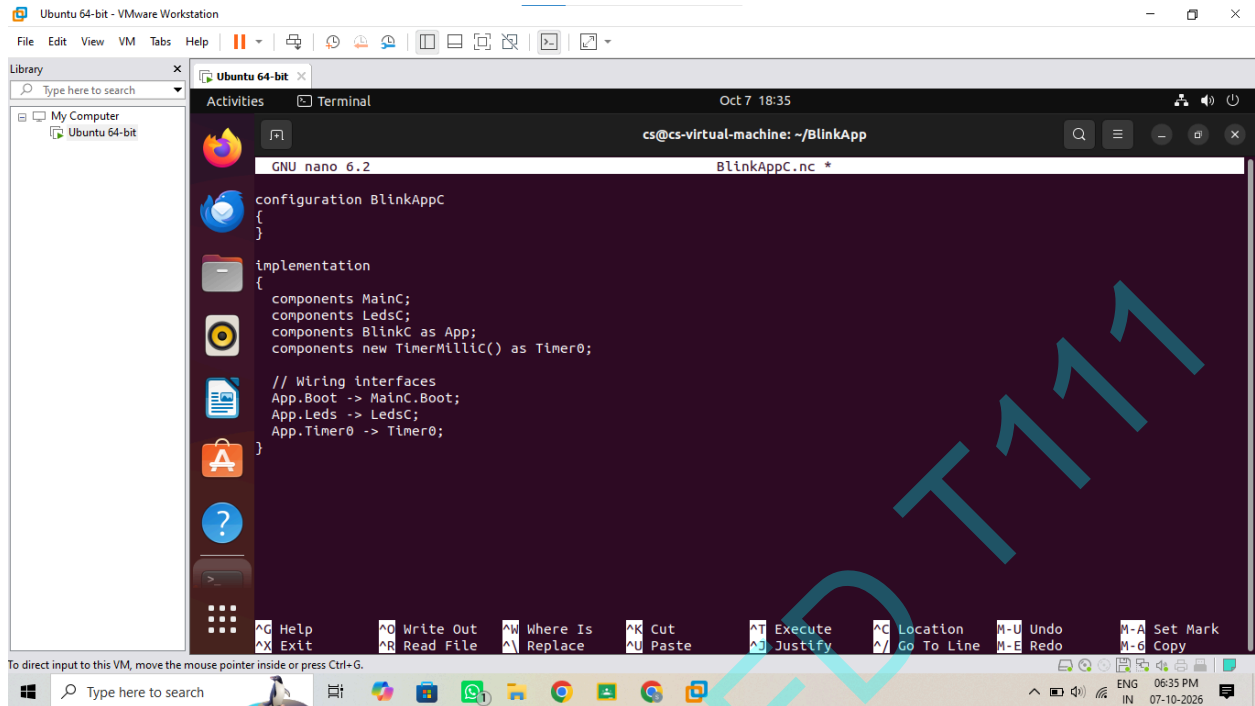
// EVENT: Timer fired
event void Timer0.fired()
{
    dbg("BlinkC",
        "EVENT: Timer0 fired! Posting task to TinyOS scheduler...\n");

    // Post task for deferred execution
    post toggleLedTask();
}
}
```

Create the configuration file:

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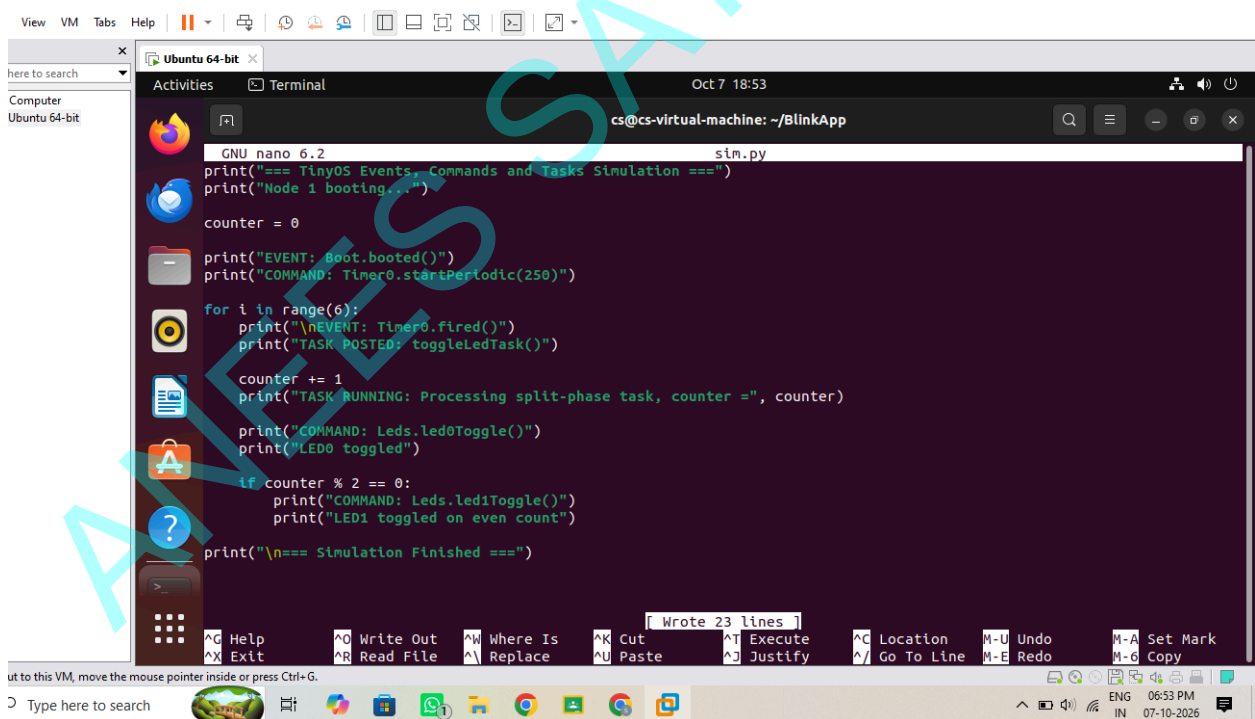


```
configuration BlinkAppC
{
}

implementation
{
    components MainC;
    components LedsC;
    components BlinkC as App;
    components new TimerMillic() as Timer0;

    // Wiring interfaces
    App.Boot -> MainC.Boot;
    App.Leds -> LedsC;
    App.Timer0 -> Timer0;
}
```

Create the TOSSIM Python simulation:



```
print("=== TinyOS Events, Commands and Tasks Simulation ===")
print("Node 1 booting...")

counter = 0

print("EVENT: Boot.booted()")
print("COMMAND: Timer0.startPeriodic(250)")

for i in range(6):
    print("\nEVENT: Timer0.fired()")
    print("TASK POSTED: toggleLedTask()")

    counter += 1
    print("TASK RUNNING: Processing split-phase task, counter =", counter)

    print("COMMAND: Leds.led0Toggle()")
    print("LED0 toggled")

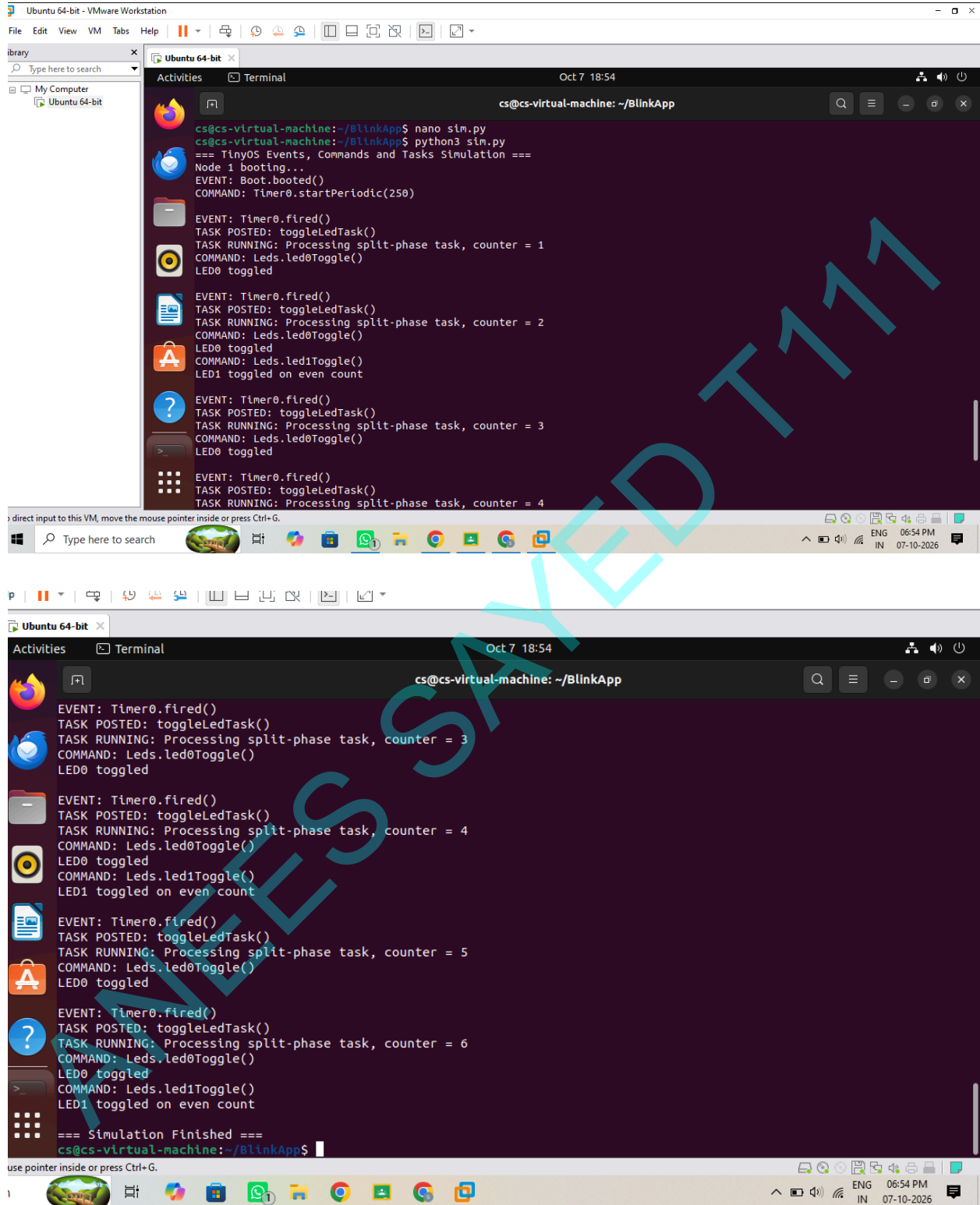
    if counter % 2 == 0:
        print("COMMAND: Leds.led1Toggle()")
        print("LED1 toggled on even count")

print("\n=== Simulation Finished ===")
```

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Activities Terminal

Oct 7 18:54

cs@cs-virtual-machine: ~/BlinkApp

```
cs@cs-virtual-machine:~/BlinkApp$ nano sim.py
cs@cs-virtual-machine:~/BlinkApp$ python3 sim.py
=== TinyOS Events, Commands and Tasks Simulation ===
Node 1 booting...
EVENT: Boot.booted()
COMMAND: Timer0.startPeriodic(250)

EVENT: Timer0.fired()
TASK POSTED: toggleLedTask()
TASK RUNNING: Processing split-phase task, counter = 1
COMMAND: Leds.led0Toggle()
LED0 toggled

EVENT: Timer0.fired()
TASK POSTED: toggleLedTask()
TASK RUNNING: Processing split-phase task, counter = 2
COMMAND: Leds.led0Toggle()
LED0 toggled
COMMAND: Leds.led1Toggle()
LED1 toggled on even count

EVENT: Timer0.fired()
TASK POSTED: toggleLedTask()
TASK RUNNING: Processing split-phase task, counter = 3
COMMAND: Leds.led0Toggle()
LED0 toggled

EVENT: Timer0.fired()
TASK POSTED: toggleLedTask()
TASK RUNNING: Processing split-phase task, counter = 4

EVENT: Timer0.fired()
TASK POSTED: toggleLedTask()
TASK RUNNING: Processing split-phase task, counter = 3
LED0 toggled

EVENT: Timer0.fired()
TASK POSTED: toggleLedTask()
TASK RUNNING: Processing split-phase task, counter = 4
COMMAND: Leds.led0Toggle()
LED0 toggled
COMMAND: Leds.led1Toggle()
LED1 toggled on even count

EVENT: Timer0.fired()
TASK POSTED: toggleLedTask()
TASK RUNNING: Processing split-phase task, counter = 5
COMMAND: Leds.led0Toggle()
LED0 toggled

EVENT: Timer0.fired()
TASK POSTED: toggleLedTask()
TASK RUNNING: Processing split-phase task, counter = 6
COMMAND: Leds.led0Toggle()
LED0 toggled
COMMAND: Leds.led1Toggle()
LED1 toggled on even count

=== Simulation Finished ===
cs@cs-virtual-machine:~/BlinkApp$
```

direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Type here to search

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use pointer inside or press Ctrl+G.

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